SURYANSH KUMAR

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LinkedIn | GitHub | Portfolio

EDUCATION

VIT Bhopal University 2023 – 2027
B.Tech, Computer Science and Engineering (Core) — Bhopal, Madhya Pradesh
Relevant coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Computer Networks, Object-Oriented Programming, Software Engineering

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, SQL
Backend & Web: Node.js, Express.js, Flask, FastAPI, REST APIs, React
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Prisma ORM
AI & Machine Learning: PyTorch, convolutional neural networks (CNNs), transfer learning, Retrieval-Augmented Generation (RAG), FAISS, LangChain, Ollama, scikit-learn, Pandas, NumPy
Tools & Practices: Git/GitHub, Docker, GitHub Actions (CI), PyTest, Linux, AWS (EC2, Lambda)

EXPERIENCE

DRDO (Defence Research and Development Organisation) — Summer Trainee Apr 2026 – Jun 2026
Multimodal RAG Document Assistant · GitHub

- Built an offline document question-answering service in Python and Flask that answers questions over uploaded PDFs and cites the source page behind each answer.
- Implemented a multimodal ingestion pipeline extracting text (PyMuPDF), tables (pdfplumber) and per-page image descriptions from a local vision model, extending retrieval to charts and scanned figures.
- Integrated three locally hosted models via Ollama — Llama 3 for generation, nomic-embed-text for embeddings, LLaVA for vision — against a per-document FAISS index, keeping documents on-device and removing per-query API cost.
- Implemented session-based authentication, role-based access control and an admin view on SQLite for multi-user access and per-document index management.

DRDO — Winter Trainee Nov 2025 – Jan 2026
OSINT Investigation Platform · GitHub

- Built a full-stack investigation tool in Node.js, Express and React that extracts named entities from uploaded documents and file metadata.
- Integrated a locally hosted Llama 3 model via Ollama into an iterative investigation pipeline for keyword extraction and draft report generation.
- Designed a normalized PostgreSQL schema covering 6+ entity types with Prisma ORM to support relationship mapping across cases.
- Wrote 5+ Python automation scripts to collect and enrich information from public web sources.

PROJECTS

Malware Family Classifier — Deep Learning for Static Analysis · GitHub · Python, PyTorch, FastAPI, Docker

- Trained a 250K-parameter CNN from scratch in PyTorch to classify malware binaries into 25 families from grayscale byte-image representations (0.939 validation macro-F1), and benchmarked it against a fine-tuned ResNet-18 at 11.2M parameters (0.985), a 45× larger model.
- Selected macro-F1 and per-family recall over accuracy after profiling a 37× class imbalance (2,949 to 80 samples per family), applying inverse-square-root class weighting in the loss to keep rare families learnable.
- Evaluated the closed-set assumption by withholding one family from training, then added max-softmax rejection; served the model behind a FastAPI endpoint with PyTest tests, a Docker image and GitHub Actions CI.

Mind Connect — Mental Health Support Platform · React, Node.js, Express.js, JWT

- Built a full-stack web application supporting anonymous posting and community discussion.
- Implemented JWT session authentication and bcrypt password hashing (10 rounds) behind a stateless REST API.

CERTIFICATIONS & TRAINING

AWS Technical Essentials — AWS digital training · **The Bits and Bytes of Computer Networking** — Google, via Coursera · **NPTEL** — IIT Madras